

Research Notes on RMSprop optimizer

RMSprop optimizer

>> Section 1

Further proposals include the momentum method or the heavy ball method, which in ML context appeared in Rumelhart, Hinton and Williams' paper on backpropagation learning and borrowed the idea from Soviet mathematician Boris Polyak's 1964 article on solving functional equations. Stochastic gradient descent with momentum remembers the update Δw at each iteration, and determines the next update as a linear combination of the gradient and the previous update:

>> Section 2

where, γ $\{\displaystyle \gamma\}$ is the forgetting factor. The concept of storing the historical gradient as sum of squares is borrowed from Adagrad, but "forgetting" is introduced to solve Adagrad's diminishing learning rates in non-convex problems by gradually decreasing the influence of old data. And the parameters are updated as,

>> Section 3

$\max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - g(w_{k\eta})| \leq C\eta$, $\{\displaystyle \max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - g(w_{k\eta})| \leq C\eta\}$

>> Section 4

Extensions and variants Many improvements on the basic stochastic gradient descent algorithm have been proposed and used. In particular, in machine learning, the need to set a learning rate (step size) has been recognized as problematic. Setting this parameter too high can cause the algorithm to diverge; setting it too low makes it slow to converge. A conceptually simple extension of stochastic gradient descent makes the learning rate a decreasing function η_t of the iteration number t , giving a learning rate schedule, so that the first iterations cause large changes in the parameters, while the later ones do only fine-tuning. Such schedules have been known since the work of MacQueen on k-means clustering. Practical guidance on choosing the step size in several variants of SGD is given by Spall.

>> Section 5

Averaging Averaged stochastic gradient descent, invented independently by Ruppert and Polyak in the late 1980s, is ordinary stochastic gradient descent that records an average of its parameter vector over time. That is, the update is the same as for ordinary stochastic gradient descent, but the algorithm also keeps track of

>> Section 6

$w_{\text{new}} := \arg \min_w \{Q_i(w) + \frac{1}{2}\eta \|w - w_{\text{old}}\|^2\}$. $\{\displaystyle w^{\text{new}} := \arg \min_w \{Q_i(w) + \frac{1}{2}\eta \|w - w_{\text{old}}\|^2\}$

$\frac{1}{\sqrt{Q_i(w) + \frac{1}{2}\eta \|w - w_{\text{old}}\|^2}}$

>> Section 7

Notable applications Stochastic gradient descent is a popular algorithm for training a wide range of models in machine learning, including (linear) support vector machines, logistic regression (see, e.g., Vowpal Wabbit) and graphical models. When combined with the backpropagation algorithm, it is the de facto standard algorithm for training artificial neural networks. Its use has been also reported in the Geophysics community, specifically to applications of Full Waveform Inversion (FWI). Stochastic gradient descent competes with the L-BFGS algorithm, which is also widely used. Stochastic gradient descent has been used since at least 1960 for training linear regression models, originally under the name ADALINE. Another stochastic gradient descent algorithm is the least mean squares (LMS) adaptive filter.

>> Section 8

Linear regression Suppose we want to fit a straight line $\hat{y} = w_1 + w_2 x$ $\{\displaystyle \hat{y} = w_1 + w_2 x\}$ to a training set with observations $((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$ $\{\displaystyle ((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))\}$ and corresponding estimated responses $(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ $\{\displaystyle (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)\}$ using least squares. The objective function to be minimized is

>> Section 9

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

>> Section 10

where $x_j' w = x_{j1} w_1 + x_{j2} w_2 + \dots + x_{jp} w_p$ $\{\displaystyle x_j' w = x_{j1} w_1 + x_{j2} w_2 + \dots + x_{jp} w_p\}$ indicates the inner product. Note that x $\{\displaystyle x\}$ could have "1" as the first element to include an intercept. Classical stochastic gradient descent proceeds as follows:

>> Section 11

Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a

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"second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss

>> Section 12

$\max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - E[g(W_{k\eta})]| \leq C\eta^2.$ $\{\displaystyle \max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - E[g(W_{k\eta})]| \leq C\eta^2.\}$

>> Section 13

$Q(w) = \frac{1}{n} \sum_{i=1}^n Q_i(w) = \frac{1}{n} \sum_{i=1}^n (m(w; x_i) - y_i)^2,$ $\{\displaystyle Q(w) = \frac{1}{n} \sum_{i=1}^n Q_i(w) = \frac{1}{n} \sum_{i=1}^n (m(w; x_i) - y_i)^2,\}$ where $m(w; x_i)$ $\{\displaystyle m(w; x_i)\}$ is the predictive model (e.g., a deep neural network) the objective's structure can be exploited to estimate 2nd order information using gradients only. The resulting methods are simple and often effective